PRANJAL MISHRA

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PROFILE

Robotics and Perception Engineer specializing in 3D perception, SLAM, multi-sensor calibration and fusion, and real-world robotic deployment. Experienced in building end-to-end autonomy pipelines, from LiDAR/camera/IMU sensing and state estimation to object tracking and collision avoidance, deployed on embedded platforms such as NVIDIA Jetson. Proficient in C++, Python, and ROS 1/2, with a strong track record of turning advanced algorithms into robust systems on physical hardware. Available to start immediately.

EDUCATION

Master of Science | ETH Zürich — Sep 2023 - Aug 2026 | GPA: 5.4 / 6.0
Computational Science & Engineering (Specialization: Robotics & Computer Vision)

Bachelor of Technology | Indian Institute of Technology Roorkee, India — July 2017 - June 2021 | GPA: 8.8/10
Major: Metallurgical & Materials Engineering — Rank 3
Minor: Computer Science & Engineering

TECHNICAL SKILLS

Perception & 3D Vision: Stereo/RGB-D, Point-Cloud Processing, ICP, SLAM, Multi-View Geometry, 3D Reconstruction **Calibratio**
Optimization & Mapping: Ceres Solver, GTSAM, g2o, OctoMap
Programming & Tools: C/C++, Python, CUDA, Bash, Linux, CMake, Git, Docker, CI/CD, GoogleTest, pytest, Profiling
Robotics & Simulation: ROS 1/2, Nav2, MoveIt, RViz, Foxglove, rosbag, NVIDIA Jetson, Isaac Sim, Gazebo, Unity, CARLA
Libraries & ML: OpenCV, Eigen, PCL, Open3D, NumPy, SciPy, PyTorch, YOLO, MMDetection3D
Deployment: TensorRT, ONNX, Model Quantization, Real-Time Optimization
Web & APIs: FastAPI, React, REST APIs

PROFESSIONAL EXPERIENCE

Research Assistant (Robotic Systems) | D-HEST, ETH Zürich — Aug 2024 – Aug 2026
SCAI Lab – Dr. Diego Paez-Granados — Zurich, CH

- Built and deployed real-time perception-to-control pipelines in ROS 2 (C++/Python) on NVIDIA Jetson edge hardware, including sensor calibration, multi-sensor fusion, and reproducible Docker-based deployment across robotic platforms
- Contributed to **ROS4HC (ROS for Healthcare)**, an open-source framework standardizing physiological/biological sensor integration into ROS 2 systems, combining multimodal sensor drivers, signal processing, and real-time visualization on assistive robotic platforms [GitHub]
- Led markerless motion capture and data processing for pediatric gait disorder research with ETH AIT Lab, owning the multi-view 3D reconstruction pipeline and its clinical-scale data workflows

Simulation Engineer | Research & Development — July 2021 – Aug 2023
Sterlite Technologies Limited — Aurangabad, India

- Built finite-element multiphase flow and FSI models (COMSOL) to analyse failure modes in an industrial process
- Validated models against production data and ran sensitivity studies to identify key process variables
- Turned results into process changes with operations teams; work contributed to two patents

PROJECTS

Multi-Camera 3D Human Pose & Body Model Reconstruction Pipeline — Jan 2026 - Aug 2026
(SCAI Lab × AIT Lab), ETH Zürich

- Built a multi-view perception pipeline reconstructing 3D body pose and shape from synchronized RGB/IR stereo point clouds: camera calibration, 2D keypoint detection, multi-view triangulation, and SMPL-X mesh fitting with ICP-based scan-to-mesh validation

- Developed a segmentation-driven subject isolation stage (instance segmentation → cross-view mask matching → SAM2-guided refinement) to fit body models robustly in cluttered clinical scenes with multiple people
- Engineered scalable batch workflows to run the pipeline reproducibly over large-scale clinical recordings spanning many subjects and sessions, plus a real-time visualizer with a FastAPI backend serving fitted gait body model and ground-reaction-force (GRF) data in temporal sync to a web-based frontend for biomechanical review

ArtiTwinSplat | Interactable Digital Twin Reconstruction from RGB-D Videos (Master's Thesis) June 2025 - Jan 2026
CVG Lab, ETH Zürich

- Designed a photorealistic, interactable 3D scene reconstruction pipeline based on 3D Gaussian Splatting (3DGS), extended to dynamic articulated scenes via disentangled background/object Gaussians
- Applied 4D RANSAC-based mobility fitting and an unsupervised RGB-D reconstruction framework to extract joint articulation priors for object motion estimation
- Extended Nerfstudio's WebRTC-based viewer to support live rendering of dynamic, articulated Gaussian Splat scenes, enabling real-time remote viewing of the reconstructed digital twin

ROS 2 Real-Time Perception-to-Control for Non-Holonomic Wheelchairs Feb 2025 - Nov 2025
SCAI Lab, ETH Zürich

- Built a real-time perception-to-control pipeline (C++/ROS 2) on NVIDIA Jetson, integrating dual Livox LiDARs, IMU-based state estimation (FAST-LIO), and multi-LiDAR point-cloud fusion/preprocessing
- Developed reactive crowd-navigation and collision-avoidance stack (convex bounding-shape constraints) for powered wheelchairs, including ROS 2 teleoperation interfaces for manual control and on-vehicle testing [GitHub]
- Integrated pedestrian detection (M-detector) with RVO2-based reactive collision avoidance for safe navigation in dynamic multi-agent environments

PokeSDF | A Unified Multi-Modal Representation for Deformable Objects (Semester Project) Nov 2024 - Feb 2025
SRL Lab, ETH Zürich

- Built an SDF-based 3D reconstruction pipeline fusing vision, force, and end-effector pose to infer object deformation from poking interactions, using a template-free approach [Project]
- Performed hand-eye calibration to register the camera frame to the robot's end-effector for accurate multi-modal fusion

AR Room Layout Estimation (Mixed Reality Course Project) Sept 2024 - Dec 2024
CVG Lab, ETH Zürich

- Built a real-time AR room-layout estimation application on Magic Leap using spatial mapping for boundary and object placement

Grasp Motion Diffusion Model (GMDM) (Digital Humans Course Project) Feb 2024 - May 2024
VLG Lab, ETH Zürich

- Developed a conditional latent diffusion model for single-stage human grasping motion generation, demonstrated on motion infilling and object trajectory prediction [Project]

PUBLICATIONS & PATENTS

1. 2026. **P. Mishra**, R. Zurbrügg, M. Wilder-Smith, M. Hutter, M. Pollefeys, Z. Bauer.
ArtiTwinSplat: Interactable Digital Twin Reconstruction via Gaussian Splatting from RGB-D Videos.
ICRA 2026 Workshop on Advances and Challenges in AI-Driven Automation and Robotic System Integration with Digital Twins, Vienna, 2026. Lightning Talk & Poster. arXiv

2. 2026. R. J. Manríquez-Cisterna†, **P. Mishra**†, J. Pena-Queralta, M. Perez-Serrano, S. Garyfallidis, L. Kupper, M. Ejtehadi, A. Breuss, A. A. Ravankar, J. V. Salazar Luces, R. Riener, Y. Hirata, D. Paez-Granados.
ROS 4 Healthcare: Physiological Human Sensing for Social, Assistive, Rehabilitation, and Medical Robotics.
Frontiers in Robotics and AI, 2026. DOI: 10.3389/frobt.2026.1745197

3. Two prior patents (2022–2023) and one materials-science publication (2021) from earlier engineering work – available on request.